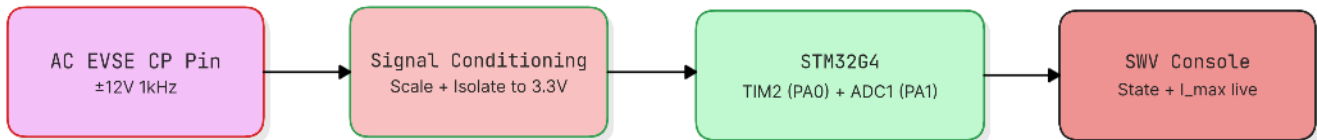
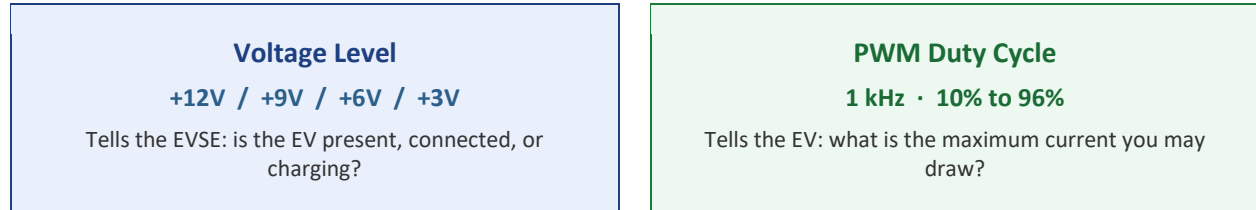


STM32 EVSE CP Signal Analyzer

Signal Flow in CP

In standard AC EV charging, the EVSE and the EV communicate entirely through the **Control Pilot (CP)** signal, governed by the **IEC 61851-1** protocol. The CP signal carries two independent pieces of information:

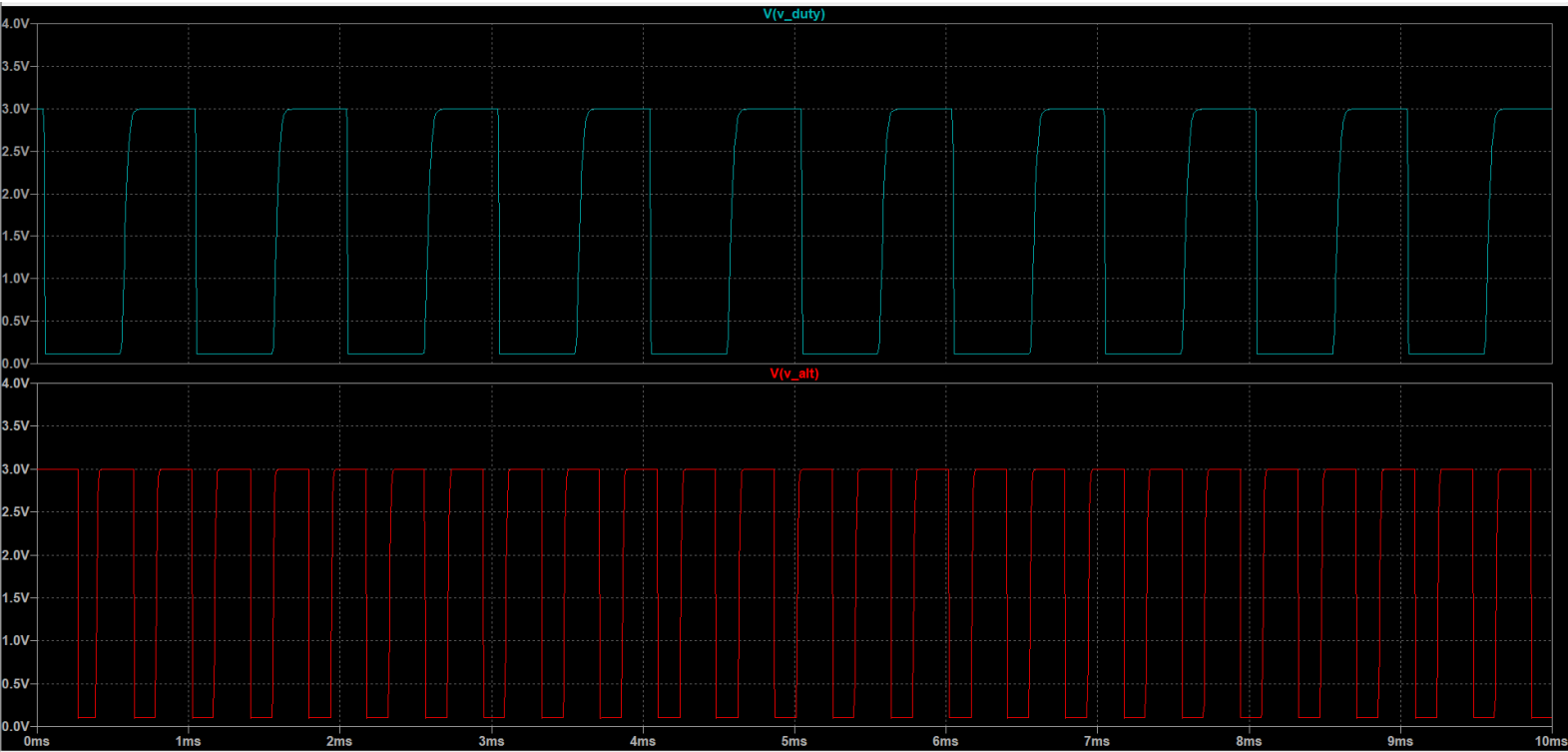
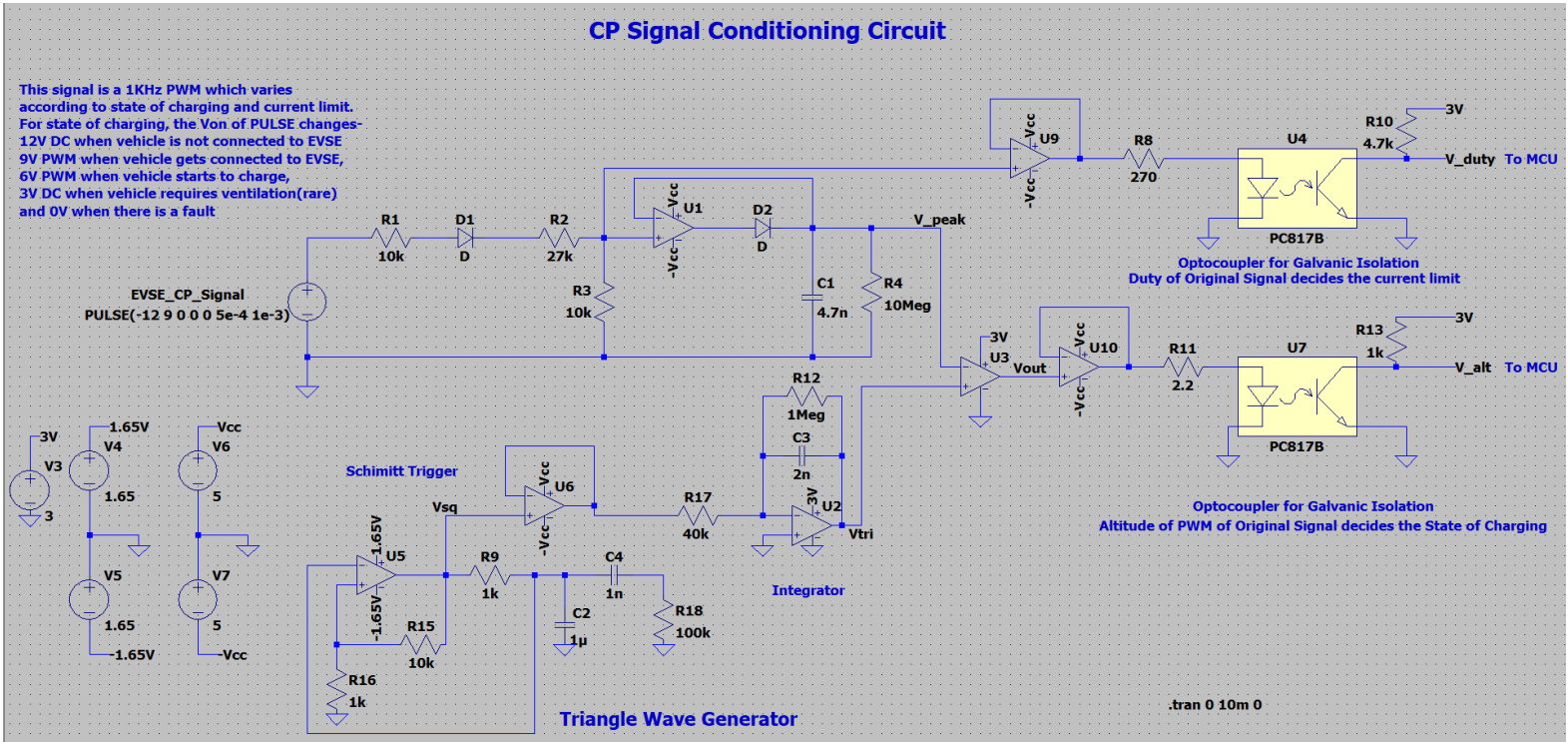


Signal Conditioning Circuit

The signal conditioning circuit scales the $\pm 12\text{V}$ CP signal to a 0–3.3V range and provides galvanic isolation (PC817 optocoupler, 5000V rated) between the EVSE mains-referenced PE ground and the MCU signal ground.

- 1) Duty of Original Signal decides the current limit.
- 2) Altitude of PWM of Original Signal decides the State of Charging.

LTSpice Simulation of signal conditioning circuit-



EV Connection States & Coded Logic

State	Description	PWM on PA0?	ADC Threshold (PA1)
A	Standby — No EV connected	No (DC only)	$ADC \geq 2600\text{ mV}$
B	EV Connected — Ready to charge	Yes (1 kHz)	$ADC \geq 1837\text{ mV}$
C	Charging — EV drawing current	Yes (1 kHz)	$400\text{ mV} \leq ADC < 1837\text{ mV}$
D	Charging + Ventilation required	No (DC only)	$400\text{ mV} \leq ADC < 2600\text{ mV}$
E	Fault — CP shorted or open circuit	No	$ADC < 400\text{ mV}$ (no prior PWM)
F	EVSE Internal Fault	No	$ADC < 400\text{ mV}$ (PWM was seen before)

Current Limit Calculation (IEC 61851-1)

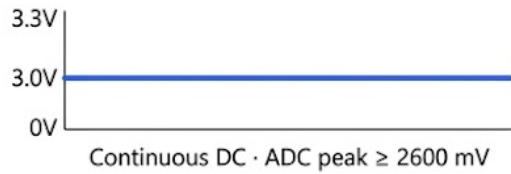
Duty Cycle	Max Current	Formula
8% to 85%	4.8 A to 51.0 A	$I = \text{duty} \times 0.6$
85% to 97%	52.5 A to 82.5 A	$I = (\text{duty} - 64) \times 2.5$
< 8% or > 97%	0 A (Fault zone)	Current = 0 (system stops)

Examples: 25% $\rightarrow 25 \times 0.6 = 15.0\text{A}$ 50% $\rightarrow 50 \times 0.6 = 30.0\text{A}$ 90% $\rightarrow (90 - 64) \times 2.5 = 65.0\text{A}$
--

Conditioned Signal Waveforms (At STM32 Input)

These waveforms are supplied directly by the function generator to PA0 and PA1, bypassing the conditioning circuit, to verify the firmware before connecting to a real EVSE.

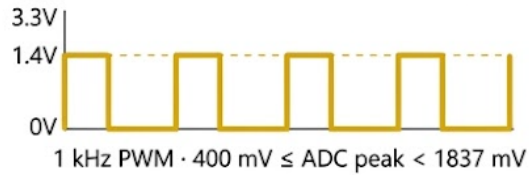
State A — No EV connected



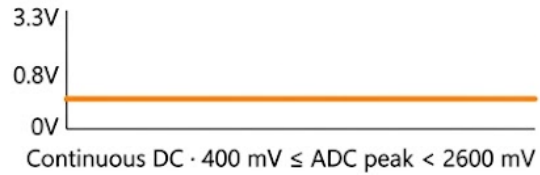
State B — EV connected



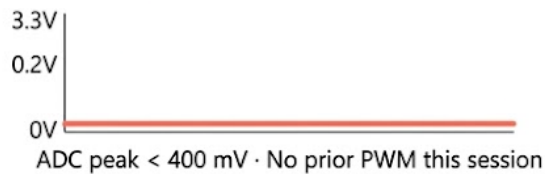
State C — Charging



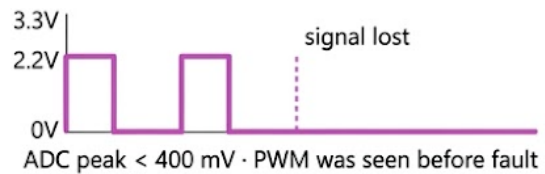
State D — Ventilation required



State E — Fault (CP shorted)



State F — EVSE fault



How the firmware decides

— State A — DC high, no PWM edges

— State D — DC low, no PWM edges

— State B — PWM present, ADC ≥ 1837 mV

— State E — 0V, PWM never seen

— State C — PWM present, ADC 400–1837 mV

— State F — 0V, PWM was seen before

All waveforms shown at the STM32G4 pins (PA0/PA1) after signal conditioning. Max 3.3V.